Open problems:
Quantum pure states are not enough

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

Theorem 5 is verified, and Theorem 3 for $\dim \mathcal{H} \geq 2$. Theorem 1 is not: see the two items marked below.

From the brief

1. [BLOCKS VERIFICATION: BRIEF ISSUE] **The range of Theorem 1.** The theorem claims consistency for $1 \leq \dim V < \infty$, while its proof uses $2 \leq \dim X$. Making it precise needs four choices.

   - *Dimension* 0. The range starts at 1 if a 0-dimensional symplectic manifold ($n = 0$ degrees of freedom) is allowed, and at 2 if not. For $\dim V = 1$ the state space is a single point, and $P(V)$ has two points exactly when $2 \leq \dim V$ (`subsingleton_projectivization`, `nontrivial_projectivization_iff`).
   - *Theorem 3 depends on it.* If a point is allowed, Theorem 3 is false for $\dim \mathcal{H} = 1$: the CST ensembles of a point and the density operators of $\mathbb{C}$ are both a single point, and the constant map identifies them (`cst_qeDens_consistent_of_finrank_one`). So the brief cannot keep both $1 \leq \dim V$ in Theorem 1 and Theorem 3 for every $\mathcal{H}$.
   - *The empty space.* For $\dim V = 0$, $P(V)$ is empty (`isEmpty_projectivization`), which is a symplectic manifold of every dimension. Either lower bound, 1 or 2, needs CST to exclude it, for instance by requiring an ensemble.
   - *What "represented by" keeps.* If X and $P(V)$ need only be in bijection, the "only if" is false: $P(\ell^2)$ and the sphere $P(\mathbb{C}^2)$ both have $\mathfrak{c}$ points (`nonempty_projectivization_equiv`), and the sphere's symplectic structure can be carried over. The theorem needs V Hausdorff and X to carry the quotient topology of $P(V)$; finiteness then comes from CST's Liouville measure ω^n with finite n, not from X being symplectic (Log, "Problems found").

   Proposed: exclude dimension 0 and the empty space, so that Theorem 1 reads $2 \leq \dim V < \infty$ for a Hausdorff V with $X = P(V)$ as topological spaces. Theorem 3 and its Lean proof then stand as they are. The alternative keeps $1 \leq \dim V$, also excludes the empty space and restricts Theorem 3 to $\dim \mathcal{H} \geq 2$. (Log, "Problems found", first three items.)

2. **Ensembles supported on a subspace.** The proof of Theorem 3 compares the ensembles with support in a two-dimensional subspace A. For $\dim \mathcal{H} \geq 3$ no CST ensemble has support

in $P(A)$, which has Liouville measure zero. Whether the brief means the classical ensembles of the subsystem $P(A)$, with its own Liouville measure, is open. The formalization compares the whole ensemble spaces instead. (Log, "Problems found", second item.)

3. **Hamiltonian evolution on the classical qubit.** Example 2 identifies Hamiltonian evolution with the diffeomorphisms that conserve the volume. Area-preserving reflections of the sphere are not Hamiltonian, so the area form should be meant. The claim that quantum evolutions are exactly the rotations is not checked. (Log, "Problems found", fourth item.)

From the formalization

1. [BLOCKS VERIFICATION: FORMALIZATION GAP] **Theorem 1.** Mathlib has no symplectic manifolds and no manifold structure on projective space, so neither direction of Theorem 1 is formalized: that $P(V)$ with the Fubini–Study form is symplectic for $\dim V < \infty$, and that a symplectic $P(V)$ forces $\dim V < \infty$. Theorem 3 for $\dim \mathcal{H} = \infty$ rests on the second direction, so it is formalized only for $\dim \mathcal{H} < \infty$. The second direction needs the topology of $P(V)$; as sets it is false (`nonempty_projectivization_equiv`). (Log, "Results".)

2. **Example 2.** The identification of $P(\mathbb{C}^2)$ with the sphere, with Darboux coordinates z, θ_{xy} and the area as Liouville measure, is not formalized. `ForPhyslib/QuantumMechanics/Qubit/Bloch.lean` shows that the Bloch vector determines the ray, which is injectivity of the map to the sphere; surjectivity and the measure are missing. (Log, "Differences", last item.)

3. **Extreme points.** The proof of Theorem 5 starts from the fact that the pure states are the extreme points of $D(\mathcal{H})$. The formalization does not need it and does not prove it. (Log, "Differences", last item.)